Question 3: Insights and Use Cases

Assignment

> On insights and use cases:
>
> - What first impressions do you get from this data?
> - Name three use cases for this dataset (assume you might also combine it with other public or Adevinta datasets).

In [1]:

```
import importlib
import sys
from pathlib import Path

MODULE_DIR = Path("modules").resolve()
if not MODULE_DIR.exists():
    MODULE_DIR = Path("assignments/cars/modules").resolve()
if str(MODULE_DIR) not in sys.path:
    sys.path.insert(0, str(MODULE_DIR))

from IPython.display import display

import functions as helpers

helpers = importlib.reload(helpers)
pd = helpers.pd

pd.set_option("display.float_format", "{:,.3f}".format)
pd.set_option("display.max_columns", 80)
helpers.set_plot_style()
```

In [2]:

```
raw_data = helpers.read_ab_test_data(helpers.DATA_PATH)
insight_data = helpers.prepare_insight_data(raw_data)
```

1. First Impressions

- The dataset captures **buyer interest in car ads**, not final sales.
- Lead activity varies by lead type.
- Lead volume is skewed across ads.

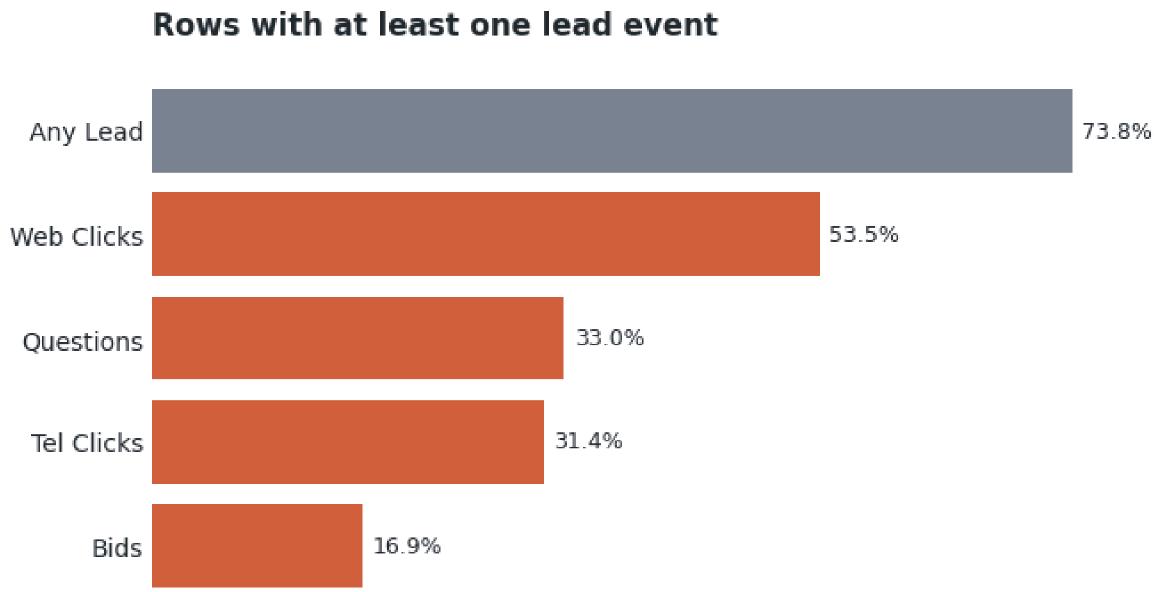
1.1 Lead Mix by Type

In [3]:

```
channel_mix = helpers.lead_channel_mix(insight_data)

display(helpers.style_table(channel_mix))
helpers.plot_lead_channel_coverage(insight_data)
helpers.plt.show()
```

	Lead Type	Rows With Event	Ads With Event Share	Avg Events/Ad	Lead Event Share
3	Web Clicks	97,931	0.535	2.739	0.514
2	Questions	60,485	0.330	0.991	0.186
0	Tel Clicks	57,429	0.314	0.917	0.172
1	Bids	30,908	0.169	0.678	0.127



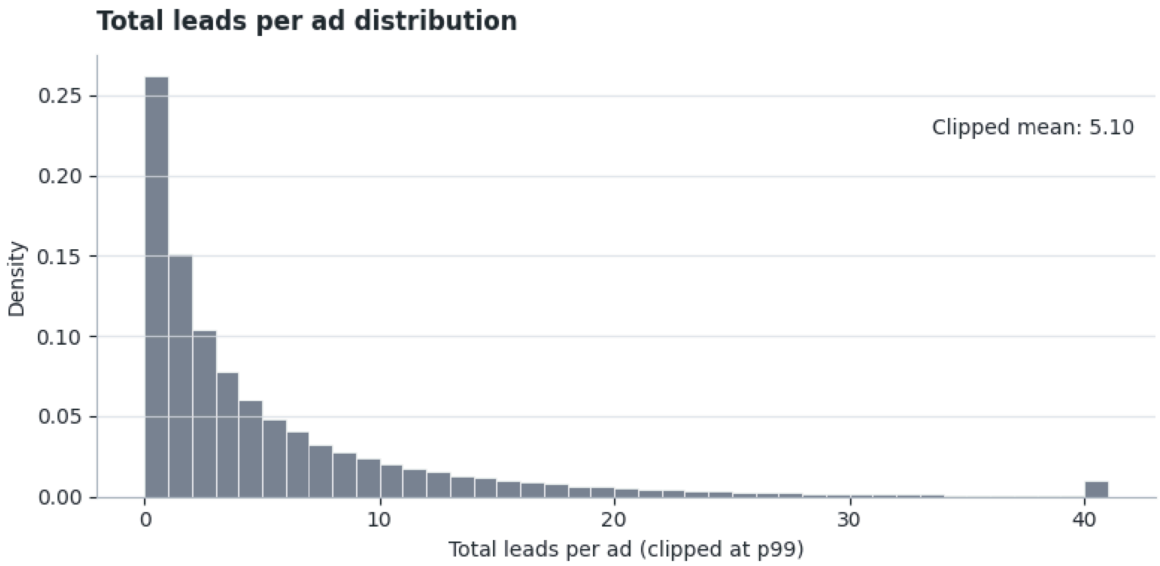
1.2 Lead Activity Is Skewed

- 74% of ads receive at least one lead.
- The median ad receives only 2 total leads.
- The top decile of ads captures about 49% of all lead events.

In [4]:

```
display(helpers.style_table(helpers.lead_concentration_summary(insight_da
helpers.plot_total_leads_distribution(insight_data)
helpers.plt.show()
```

	Metric	Value
0	Ads With At Least One Lead	0.738
1	Median Total Leads	2
2	P90 Total Leads	14
3	P99 Total Leads	40
4	Lead Events From Top-Decile Ads	0.487



2. Use Cases

2.1 Vehicle Segment Benchmarks

- Goal: identify listings that underperform similar vehicles and recommend the right seller action.
- Why it matters: marketplace averages can make healthy listings look weak, or weak listings look normal, if segment differences are ignored.
- Required join: seller/account data to turn listing-level benchmarks into seller actions.
- Helpful joins: final sale outcome to validate business impact, and listing edit history to measure whether seller changes improve performance.

Vehicle Demand Patterns

- Lower-price, older, and higher-mileage cars show higher lead rates in this dataset.
- These segment patterns make seller recommendations more defensible.
- Practical use: flag listings that underperform similar cars, then suggest a price check, content refresh, or promotion.
- Note: these are descriptive benchmarks; each band can still be affected by the mix of price, age, mileage, body type, and brand.
- A controlled analysis such as linear regression could help separate these effects, but that is out of scope for this project.

In [5]:
```
for segment_column in ["price_band", "km_band", "car_age_band"]:
    segment_summary = helpers.segment_benchmark_summary(insight_data, seg
    display(helpers.style_segment_insight_table(segment_summary))
```

	Price Band	Ads	Lead Rate	Avg Leads/Ad	Median Car Age	Median Mileage	Ad Share
0	0-25k	52,340	88.5%	8.689	15	197,588	28.6%
1	25k-60k	42,250	79.9%	5.815	10	162,664	23.1%
2	60k-120k	43,290	66.4%	3.524	6	113,698	23.6%
3	120k+	45,175	58.0%	2.694	3	62,129	24.7%

	Mileage Band	Ads	Lead Rate	Avg Leads/Ad	Median Price	Median Car Age	Ad Share
3	200k+	45,272	86.0%	7.167	€22,500	13	24.7%
2	150k-200k	35,863	82.5%	6.583	€39,450	11	19.6%
1	75k-150k	56,374	74.8%	5.117	€69,500	7	30.8%
0	0-75k	44,149	52.4%	2.525	€139,000	2	24.1%

	Car Age Band	Ads	Lead Rate	Avg Leads/Ad	Median Price	Median Mileage	Ad Share
3	13+	50,442	87.7%	8.113	€14,500	198,456	27.6%
2	8-12	52,457	81.5%	6.184	€44,950	166,580	28.7%
1	4-7	45,914	69.5%	3.941	€97,500	108,701	25.1%
0	0-3	34,249	47.0%	1.757	€169,950	30,212	18.7%

In [6]:

```
body_type_summary = helpers.segment_insight_summary(insight_data, "carros
brand_summary = helpers.segment_insight_summary(insight_data, "brand", mi

display(helpers.style_segment_insight_table(body_type_summary))
display(helpers.style_segment_insight_table(brand_summary))
```

	Body Type	Ads	Lead Rate	Avg Leads/Ad	Median Price	Ad Share
1	Coup	2,913	86.3%	7.863	€89,500	1.6%
5	Sedan (2/4-deurs)	15,472	84.1%	7.035	€59,500	8.5%
4	Overige carrosserie n	2,494	76.9%	5.712	€34,250	1.4%
0	Cabriolet	5,290	75.9%	4.677	€69,900	2.9%
6	Stationwagon	29,233	74.2%	4.709	€69,500	16.0%
7	Terreinwagen	10,128	72.9%	4.649	€169,400	5.5%
2	Hatchback (3/5-deurs)	79,944	71.9%	5.614	€49,500	43.7%
3	MPV	19,737	69.2%	3.879	€49,990	10.8%

	Brand	Ads	Lead Rate	Avg Leads/Ad	Median Price	Ad Share
12	AUDI	7,062	84.5%	6.658	€122,475	3.9%
37	DAEWOO	1,070	82.9%	7.456	€9,995	0.6%
40	DAIHATSU	1,052	82.5%	6.743	€19,500	0.6%
3	ALFA ROMEO	1,577	82.4%	6.674	€44,950	0.9%
187	VOLKSWAGEN	22,148	80.3%	6.862	€69,000	12.1%
20	BMW	9,259	80.1%	6.919	€119,900	5.1%
123	MERCEDES	7,809	79.6%	6.481	€124,500	4.3%
165	SEAT	4,906	79.4%	5.771	€47,125	2.7%
120	MAZDA	2,398	78.4%	5.169	€33,675	1.3%
31	CHEVROLET	1,513	76.7%	5.743	€38,500	0.8%

2.2 Listing Quality Optimization

- Goal: help sellers improve listing content that may be limiting buyer interest.
- Why it matters: listing advice should focus on fields sellers can actually change.
- Required join: seller/account data to connect recommendations to actual sellers.
- Helpful joins: final sale outcome to check whether better content helps sales, and listing edit history to measure whether content changes improve lead performance.

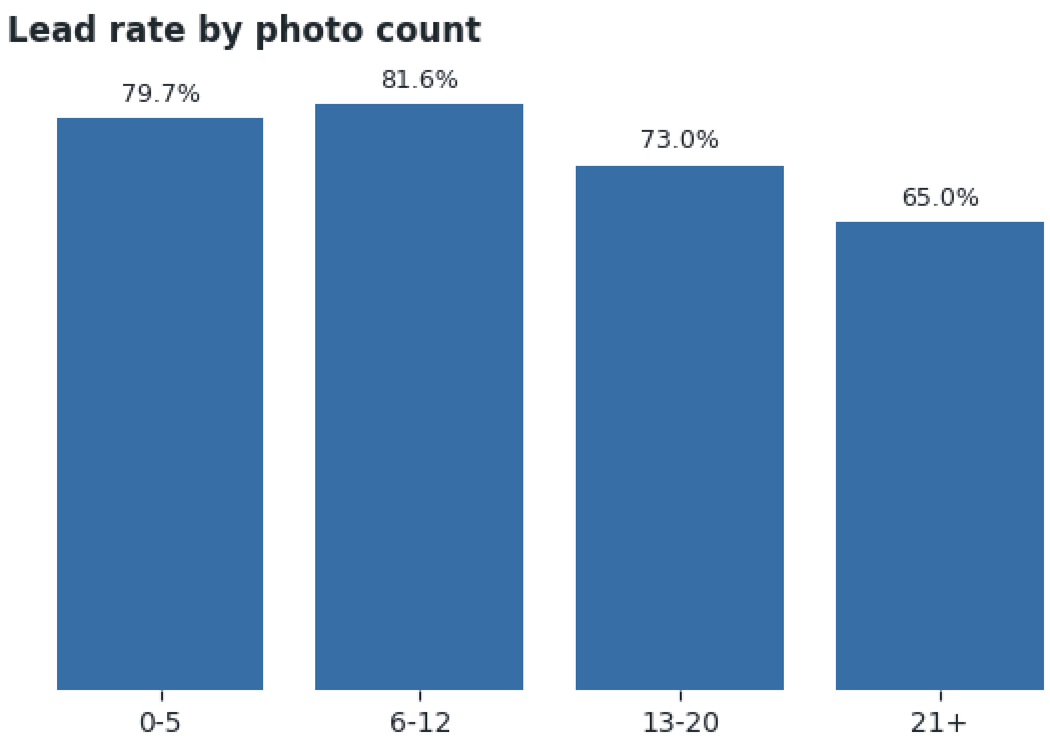
Photo Count Signals

- Photo count alone is not a simple quality score.
- High-photo listings tend to be more expensive, which also affects lead rate.
- Use photo count as a benchmark within similar vehicle and price segments.

In [7]:
```
listing_quality = helpers.listing_quality_summary(insight_data)

display(helpers.style_table(listing_quality))
helpers.plot_photo_count_lead_rate(listing_quality)
helpers.plt.show()
```

	Photo Count Band	Ads	Lead Rate	Avg Leads/Ad	Median Days Live	Median Price	Ad Share
0	0-5	8,630	79.7%	8.092	31.000	€18,500	4.7%
1	6-12	61,208	81.6%	6.968	31.000	€29,000	33.4%
2	13-20	57,750	73.0%	4.934	31.000	€59,950	31.5%
3	21+	55,474	65.0%	3.488	31.000	€117,450	30.3%



- Practical use: flag listings with unusually weak content compared with similar cars.
- Avoid a simplistic rule like "more photos always means more leads".

2.3 Listing Lifecycle Nudges

- Goal: nudge sellers at the right moment, before a listing becomes stale.
- Why it matters: listing age changes how we should read lead performance.
- Required join: seller/account data to deliver nudges to the right sellers.
- Helpful joins: listing edit history to detect seller changes, paid exposure history to separate organic from promoted performance, and final sale outcome to validate business impact.

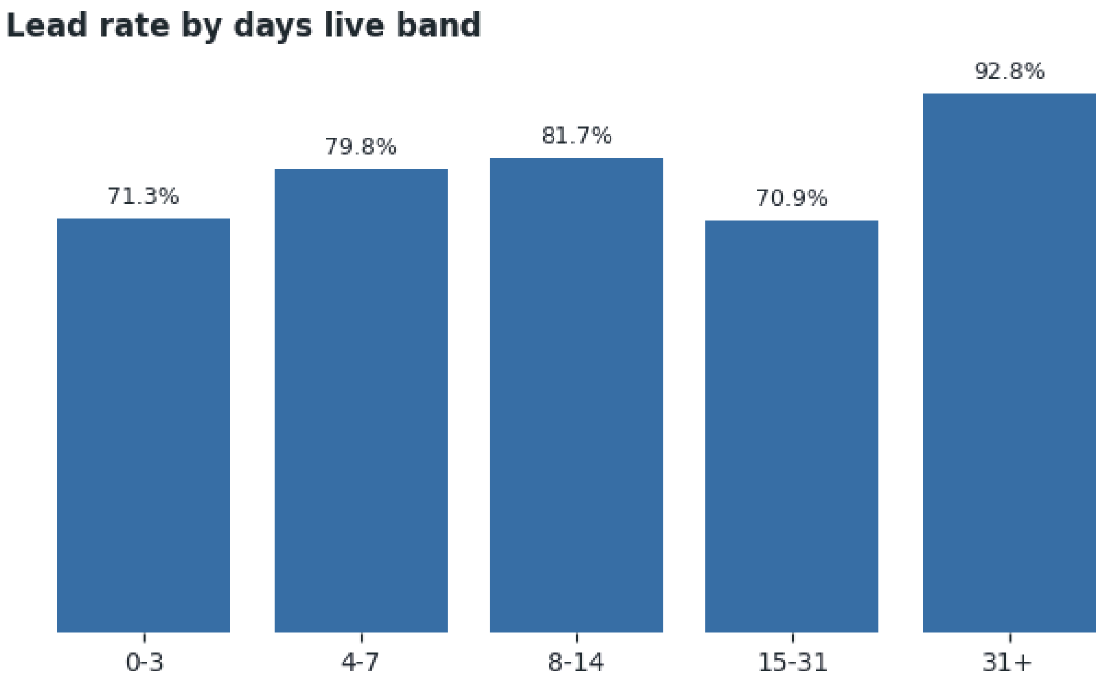
Listing Lifecycle Signals

- Check early if new listings are getting leads.
- Mid-life listings can be compared against segment benchmarks.
- Stale listings with weak lead performance may need a refresh, price check, or promotion.
- These lifecycle patterns are descriptive, not causal. I would use them as triggers for diagnosis, not as proof that listing age drives lead performance.

In [8]:
```
lifecycle_summary = helpers.lifecycle_summary(insight_data)

display(helpers.style_table(lifecycle_summary))
helpers.plot_lifecycle_lead_rate(lifecycle_summary)
helpers.plt.show()
```

	Days Live Band	Ads	Lead Rate	Avg Leads/Ad	Median Leads/Ad	Median Price	Median Photos	Ad Share
0	0-3	14,771	71.3%	5.486	3.000	€29,990	12.000	8.1%
1	4-7	14,766	79.8%	7.916	4.000	€36,500	14.000	8.1%
2	8-14	16,583	81.7%	7.199	4.000	€39,990	15.000	9.1%
3	15-31	127,195	70.9%	4.483	2.000	€69,500	16.000	69.5%
4	31+	9,747	92.8%	8.951	6.000	€37,500	11.000	5.3%



In [9]:
```
fresh_stale_summary = helpers.fresh_stale_summary(insight_data)

display(helpers.style_table(fresh_stale_summary))
```

	Lifecycle Stage	Ads	Lead Rate	Avg Leads/Ad	Median Leads/Ad	Median Price	Median Photos	Median Mileage	Ad Share
0	Fresh: 0-7 days	29,537	75.5%	6.701	3.000	€33,500	13.000	155,151	16.1%
1	Mid-life: 8-31 days	143,778	72.1%	4.796	2.000	€66,900	16.000	132,728	78.5%
2	Stale: 31+ days	9,747	92.8%	8.951	6.000	€37,500	11.000	169,400	5.3%

- Practical use: trigger different recommendations based on listing age and lead performance.
- Example: a stale listing with low leads can receive a refresh or promotion suggestion.